VF Asia Footwear Laboratory Testing Manual

Test Method for BALLY Flex	**Test method # VF-D01** **REF # : TM-R- 005** **Page 1 of 5**
Date Issued : June,2019 **Revision # : 3** **Reason for revision: Hydrolysis temperature changed from 70°C to 60°C.**	**Created by : Nila Babia Erazo**

PURPOSE AND SCOPE: To measure the resistance to cracking, delamination, and discoloration of the surface/coating of a test material when subjected to repeated flexing.

REFERENCE: DIN 53351

EQUIPMENT AND MATERIAL:

1. Bally flexometer [A], conforming to DIN 53351
2. Cold Bally flexometer [B], conforming to DIN 53351
3. Immersion dish
4. Distilled water
5. Hydrolysis oven, capable of maintaining temperature @ 60 ± 1ºC and relative humidity at 95 ± 1%.
6. Freezing chamber capable of maintaining temperature @ - 20 ± 1ºC or at the specified freezing temperature.
7. Full grain leather, Classic garment leather and PU-coated leather, fabric, and synthetic leather.

SAMPLE PREPARATION:

1. Condition the sample for at least 24 hours at the standard laboratory condition at 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.
2. Die-cut four samples (two samples in the length direction and two samples in the width direction) with length of 70mm and width of 45mm.The samples should be taken from varying locations on the test material, and at least 10 cm from any edges.

REPETITIONS: Test four (4) samples (two in each direction) per material per testing condition.

TEST PROCEDURE:

Test should be conducted at standard laboratory condition @ 23.0± 2.0˚C temperature and 55 ±10 % Relative Humidity.

A. Dry Test, 23 ± ºC

1. Bring the upper clamp of the bally flexometer to its maximum forward position. This can be done by manually turning the adjusting knob which is located along the side of the machine.

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2. Hold the test sample such that the outer (full grain of coated) surface is facing upward, Fold the lengthwise sides upwards such that the two edges are brought together, lightly crease the sample along its centerline.

3. Place the folded sample into the top clamp of the bally flexometer such that the crease lies along the bottom of the clamp, slide the sample to the back of the top clamp(until the back edge of the sample contacts the back of the top jaw),and firmly clamp the sample by tightening the screw.(see **Figure A**)

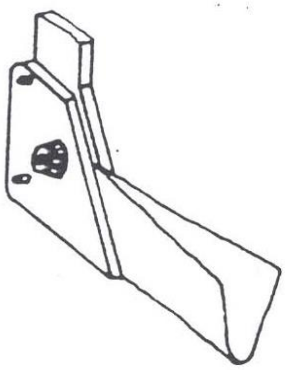


Figure A

4. The portion of the test sample which is projecting from the top clamp is then folded downwards over the top clamp so that the fold is directed vertically downwards (see **Figure B**)

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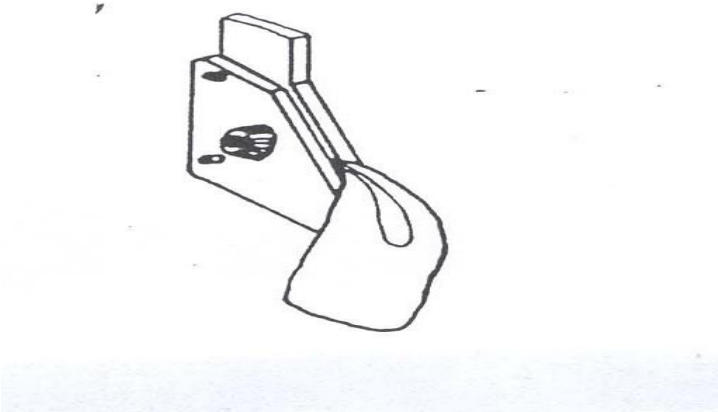


Figure B

5. Insert the free end of the test sample into the bottom clamp such that the front edge of the test sample is perfectly vertical. Lightly pinch the portion of the test sample which is located on the upper jaw (this is done to remove any excess slack in the material which may have resulted from folding the test sample over the top jaw.) Allow the lower portion of the free end to hang freely in the bottom clamp, and secure the sample by firmly tightening the screw on the bottom clamp (see **Figure C**).

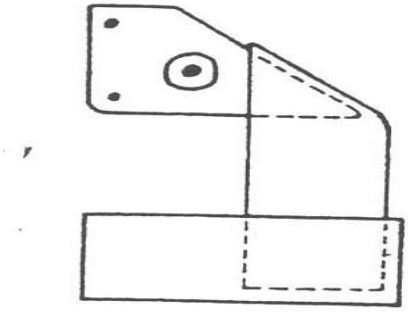


Figure C

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Note: When properly mounted, the test sample will have no tension between the top and bottom clamps.

6. Set the “shut-off “counter to the desired number of cycles, and zero the “sample” counter.

7. Initiate testing and inspect the samples after every 5000 cycles during the first 25000 cycles, after 25000 cycles inspect the samples every 10000 cycles.

8. Remove failed samples at their point of failure. If failure has not occurred after the specified # of cycles such as 100,000 cycles or 125, 000, discontinue testing. Report the result as “120,000+” or “125,000+” cycles.

Note: Failure is defined as any visible changes on the surface of the test sample which can be observed using a 6X magnifying glass, including cracking, delamination, or discoloration. For leather, failure also includes destruction of the leather fibers so that pitting, cracking or holes can be observed.

B. Wet Test, 23 ± 2ºC

1. Immerse the test samples in distilled water at 23±2 ºC for 120±5 minutes.

2. Remove the samples and wipe off any excess surface moisture.

3. Immediately, insert the samples in the bally flexometer as described in section A, **Dry Test**. Initiate testing.

4. Samples should be evaluated after every 5000 cycles. If failure has not occurred after the specified # of cycles such as 20,000 cycles or 25, 000, discontinue testing. Report the result as “20,000+” or “25,000+” cycles.

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C. Test after hydrolysis, 23 ± 2°C

1. Place the test materials in an oven at 60 ± 2°C for 7 days at 95 % relative humidity
2. After 7 days, remove the specimen from the oven.
3. Let it cool for 24 hours at 23 ± 2°C and 55 ±10% relative humidity.
4. After conditioning, insert the samples in the bally flexometer as described in section A, **Dry Test.**
5. Samples should be evaluated every 5000 cycles. If failure has not occurred after the specified # of cycles such as 50,000 cycles or 75, 000, discontinue testing. Report the result as “50,000+” or “75,000+” cycles.

D. Test at freezing temperature – 20°C or at specified freezing temperature by the brand.

Follow steps in section A, **Dry Test** .The difference is that the test is carried out in the freezing chamber.

Note: Ensure that the chamber is close and allow to reach @ - 20°C temperature (or ***at the specified freezing temperature***) before start testing.

REPORT: The number of cycles when failure occurred .